

Planet-Constrained Hydrothermal Vent Chemistry Coupled to PHREEQC and Protocell Evolution

LIFEORIG working model specification

June 14, 2026

1 Purpose

This document defines a simulation workflow in which planetary parameters determine the physical and chemical boundary conditions for an ocean–hydrothermal vent system. PHREEQC is then used to compute local aqueous speciation, pH, ionic strength, redox state, and mineral saturation. The output of PHREEQC feeds a non-equilibrium organic kinetic network that produces reduced carbon species, activated intermediates, amino-acid precursors, fatty acids, peptides, and compartment-like structures. These products are then used to define a probabilistic protocell nucleation and evolution layer.

The central pipeline is

Planet model → bulk ocean/vent chemistry → PHREEQC speciation → organic kinetic network → protocell dyn

(1)

The key modeling rule is:

PHREEQC does not create the organic network. PHREEQC provides the local chemical environment in which the organic network evolves.

PHREEQC is used as the geochemical equilibrium/speciation engine because it can calculate aqueous species distributions, saturation indices, reaction paths, and one-dimensional transport/inverse geochemical calculations [1, 2, 3].

2 Current implementation status

The current repository contains a PHREEQC smoke test in:

```
src/ocean_chem/test_ocean_chem_0.py
```

The test imports `phreeqpy`, loads:

```
external/phreeqc/database/phreeqc.dat
```

and runs a simple NaCl solution with selected output for pH, temperature, ionic strength, Na, and Cl. At the time this note was added, `phreeqpy` was available in the local project virtual environment `env/bin/python`. The conda environment used for other work may need a clean PETSc/`petsc4py`/PHREEQC setup before this backend is folded into the ODE solver stack.

3 Model hierarchy

3.1 Layer 1: planet and geophysical environment

For a water world or icy moon, define a planet object

$$\mathcal{P} = \{R, M, g, d_{\text{ocean}}, d_{\text{ice}}, \Phi_{\text{tidal}}, \Phi_{\text{rad}}, X_{\text{rock}}, X_{\text{ice}}, \mathcal{C}_{\text{rock}}\}. \quad (2)$$

The model should provide:

$$T_{\text{ocean}} = T_{\text{ocean}}(\mathcal{P}), \quad (3)$$

$$T_{\text{vent}} = T_{\text{vent}}(\Phi_{\text{tidal}}, \Phi_{\text{rad}}, \mathcal{C}_{\text{rock}}), \quad (4)$$

$$P(z) = P_0 + \rho_{\text{ocean}} g z, \quad (5)$$

$$F_{\text{H}_2} = F_{\text{H}_2}(\text{water-rock interaction}), \quad (6)$$

$$F_{\text{ox}} = F_{\text{ox}}(\text{surface radiolysis, oxidant delivery}). \quad (7)$$

The physical domain can initially be one-dimensional:

$$x = 0 \quad \text{vent boundary}, \quad x = L \quad \text{ocean boundary}. \quad (8)$$

A minimal imposed temperature profile is

$$T(x) = T_{\text{vent}} + (T_{\text{ocean}} - T_{\text{vent}}) \frac{x}{L}. \quad (9)$$

A more realistic model can solve advection–diffusion heat transport:

$$\rho c_p \left(\frac{\partial T}{\partial t} + u \frac{\partial T}{\partial x} \right) = \frac{\partial}{\partial x} \left(k_T \frac{\partial T}{\partial x} \right) + Q_{\text{local}}. \quad (10)$$

3.2 Layer 2: PHREEQC geochemistry

For each grid cell x_j , PHREEQC receives:

$$\text{PHREEQC input}(x_j) = \{T_j, P_j, \text{pH}_{\text{guess}}, \text{pe}_{\text{guess}}, \mathbf{b}_j, \mathcal{M}_j, \mathcal{G}_j\}, \quad (11)$$

where $\mathbf{b}_j$ is the vector of bulk elemental totals, $\mathcal{M}_j$ is the set of allowed mineral phases, and $\mathcal{G}_j$ is the optional gas phase.

PHREEQC returns:

$$\text{geo}_j = \{\text{pH}_j, \text{pe}_j, I_j, a_i(x_j), C_i(x_j), \text{SI}_m(x_j)\}. \quad (12)$$

Here a_i are activities, C_i are aqueous concentrations, and SI_m are mineral saturation indices.

3.3 Layer 3: organic kinetic network

The organic species evolve by reaction–transport:

$$\frac{\partial C_i^{\text{org}}}{\partial t} = -\frac{\partial}{\partial x} (u C_i^{\text{org}}) + \frac{\partial}{\partial x} \left(D_i \frac{\partial C_i^{\text{org}}}{\partial x} \right) + R_i(\mathbf{C}^{\text{org}}, \text{geo}, T, \mathcal{M}). \quad (13)$$

Each reaction rate is evaluated using PHREEQC outputs:

$$r_\alpha = k_\alpha(T) \prod_i a_i^{\nu_{i\alpha}} f_\alpha(\text{pH}) g_\alpha(I) h_\alpha(\text{minerals}) q_\alpha(\text{redox}). \quad (14)$$

The temperature-dependent prefactor may be represented as

$$k_\alpha(T) = k_{\alpha,0} \exp\left(-\frac{E_\alpha}{RT}\right). \quad (15)$$

Mineral catalysis can be represented initially as a multiplicative factor:

$$h_\alpha(\text{minerals}) = 1 + \sum_m \eta_{\alpha m} A_m \Theta_m, \quad (16)$$

where A_m is the reactive mineral surface area and Θ_m is an availability factor derived from saturation state or explicit mineral abundance.

3.4 Layer 4: protocell nucleation and evolution

The organic network produces coarse pools:

$$\mathbf{O} = \{C_{\text{formate}}, C_{\text{acetate}}, C_{\text{pyruvate}}, C_{\text{thioester}}, C_{\text{acetylP}}, C_{\text{AA}}, C_{\text{peptide}}, C_{\text{FA}}, C_{\text{vesicle}}\}. \quad (17)$$

A local protocell nucleation score is

$$S_{\text{proto}} = w_L \log(1 + C_{\text{FA}}) + w_P \log(1 + C_{\text{peptide}}) + w_E \log(1 + C_{\text{thioester}} + C_{\text{acetylP}}) + w_G \Delta\mu_{\text{redox}} + w_S S_{\text{surface}} - \Pi_{\text{stress}}. \quad (18)$$

The probability of nucleation in cell x_j during Δt is

$$P_{\text{proto}}(x_j, \Delta t) = 1 - \exp[-\lambda_0 \Delta t \sigma(S_{\text{proto}})], \quad \sigma(S) = \frac{1}{1 + e^{-S}}. \quad (19)$$

The stress penalty should include:

$$\Pi_{\text{stress}} = b_T \Pi_T + b_{\text{pH}} \Pi_{\text{pH}} + b_{\text{div}} \Pi_{\text{Mg/Ca}} + b_{\text{dil}} \Pi_{\text{dilution}} + b_{\text{hyd}} \Pi_{\text{hydrolysis}}. \quad (20)$$

4 Hydrothermal vent model structure for implementation

The hydrothermal vent model should not start by assuming free protocells. In a vent setting the first chemically useful compartments are mineral pores, surface-bound reaction domains, and porous precipitate barriers. These objects are part of the environment. They concentrate feedstocks, expose them to mineral surfaces, impose residence times, and experience continuous exchange with vent fluid and ocean water. Coacervates and vesicles should appear only as later states created by the chemistry.

The implementation hierarchy is therefore:

$$\boxed{\text{HydrothermalVent} \rightarrow \text{spatial cells} \rightarrow \text{mineral pore/surface reactors} \rightarrow \text{coacervates} \rightarrow \text{vesicles/protocell candidate}} \quad (21)$$

4.1 Top-level hydrothermal vent object

The hydrothermal vent object represents a one-dimensional vent-ocean mixing zone. It owns the environmental fields and the collection of compartment objects living inside the vent system:

$$\mathcal{V}_{\text{vent}} = \{x_j, T_j, P_j, \text{pH}_j, I_j, \text{pe}_j, u_j, \lambda_j, \mathbf{C}_j^{\text{env}}, \mathbf{S}_j^{\text{mineral}}, \mathcal{C}_j\}. \quad (22)$$

Here x_j is the spatial coordinate, u_j is the local flow velocity, λ_j is the vent/ocean mixing fraction, $\mathbf{C}_j^{\text{env}}$ are external aqueous concentrations or activities, $\mathbf{S}_j^{\text{mineral}}$ stores reactive mineral surface availability, and $\mathcal{C}_j$ is the list of compartments in cell j .

The main variables are:

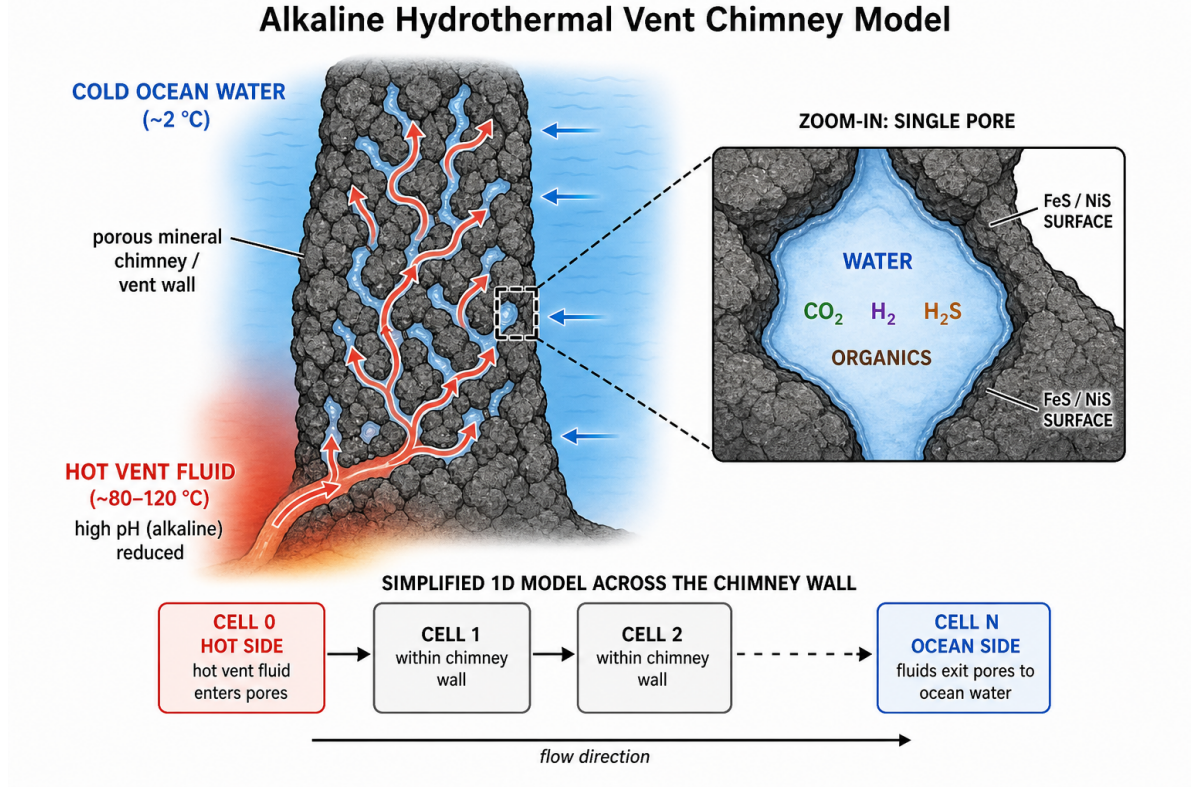


Figure 1: Coarse-grained geometry of the hydrothermal vent model. The physical object is a porous mineral chimney or vent wall separating hot reduced vent fluid from colder ocean water. The one-dimensional coordinate is an effective coordinate through this porous reactive wall, not an empty line in the ocean. Each model cell represents a slice of water-filled mineral pores with local temperature, pH, flow, dissolved chemistry, and reactive mineral surface area such as FeS or NiS. Pore and surface chemistry happen first; coacervates and vesicle-like objects are later products of that chemistry.

- x_j is the position of cell j across the porous chimney wall. The coordinate runs from hot vent side to cold ocean side.
- T_j is the temperature in cell j .
- P_j is the pressure in cell j .
- pH_j is the acidity/basicity in cell j .
- I_j is ionic strength, a measure of dissolved salt content.
- pe_j is the redox state used by PHREEQC.
- u_j is the effective flow velocity through the pores of cell j .
- λ_j is the mixing fraction: $\lambda_j = 1$ means pure vent fluid, $\lambda_j = 0$ means pure ocean water.
- $\mathbf{C}_j^{\text{env}}$ is the vector of dissolved external chemical concentrations in the pore water of cell j .

- $\mathbf{S}_j^{\text{mineral}}$ stores available mineral surface area in cell j , for example FeS, NiS, magnetite, brucite, or silica surface.
- $\mathcal{C}_j$ is the list of compartments currently inside cell j , such as coacervates or vesicle candidates.

A first implementation can use fixed profiles:

$$T(x) = T_{\text{vent}} + (T_{\text{ocean}} - T_{\text{vent}})x/L, \quad (23)$$

$$\lambda(x) = \exp(-x/\ell_{\text{mix}}), \quad (24)$$

$$C_i^{\text{env}}(x) = \lambda(x)C_{i,\text{vent}} + [1 - \lambda(x)]C_{i,\text{ocean}}. \quad (25)$$

Later, these profiles can be replaced by advection–diffusion and PHREEQC transport.

4.2 Spatial cells

Each spatial cell is a local reactor environment:

$$\mathcal{G}_j = \{T_j, P_j, \text{pH}_j, I_j, \text{pe}_j, \mathbf{a}_j, \text{SI}_{m,j}, A_{m,j}, u_j\}. \quad (26)$$

PHREEQC should provide the geochemical quantities $\mathbf{a}_j$, $\text{SI}_{m,j}$, and mineral constraints. The organic kinetic network then uses these values to compute effective rates.

4.3 Mineral pore reactors

A pore is a fixed environmental compartment, not a protocell. It does not divide and it is not inherited. It is the first hydrothermal object that can concentrate and process chemistry.

For pore k in spatial cell j :

$$\mathcal{P}_{jk} = \{x_j, r_k, L_k, V_k, A_k, \mathbf{C}_{jk}^{\text{pore}}, \mathbf{A}_{jk}^{\text{mineral}}, \tau_{jk}^{\text{res}}, q_{jk}\}. \quad (27)$$

The pore volume and surface area can initially be

$$V_k = \pi r_k^2 L_k, \quad (28)$$

$$A_k = 2\pi r_k L_k, \quad (29)$$

$$\frac{A_k}{V_k} = \frac{2}{r_k}. \quad (30)$$

The pore chemistry evolves as

$$\frac{dC_{i,k}^{\text{pore}}}{dt} = \sum_{\alpha} \nu_{i\alpha} r_{\alpha,k}^{\text{pore}} + k_{i,k}^{\text{in}} C_{i,j}^{\text{env}} - k_{i,k}^{\text{out}} C_{i,k}^{\text{pore}} - k_{i,k}^{\text{loss}} C_{i,k}^{\text{pore}}. \quad (31)$$

The reaction rate inside a pore is

$$r_{\alpha,k}^{\text{pore}} = k_{\alpha}(T_j) \prod_i (C_{i,k}^{\text{pore}})^{\nu_{i\alpha}^+} f_{\alpha}(\text{pH}_j) g_{\alpha}(I_j) h_{\alpha}(\mathbf{A}_{jk}^{\text{mineral}}) q_{\alpha}(\text{pe}_j). \quad (32)$$

The residence-time interpretation is

$$k_{i,k}^{\text{out}} \sim \frac{1}{\tau_{jk}^{\text{res}}}, \quad \tau_{jk}^{\text{res}} \sim \frac{L_k}{u_j}. \quad (33)$$

Small pores have large surface-to-volume ratio and therefore stronger surface chemistry, but high flow can flush products before autocatalytic or compartment-forming chemistry accumulates.

4.4 Surface-bound reaction domains

Surface domains are reaction states attached to minerals rather than enclosed volumes. They matter in hydrothermal vents because FeS, NiS, magnetite, brucite, and silica can retain intermediates and catalyze carbonyl, thioester, peptide, and redox chemistry.

For adsorbed species i on mineral m :

$$\frac{d\Gamma_{i,m}}{dt} = k_{i,m}^{\text{ads}} C_i^{\text{env}} (1 - \theta_m) - k_{i,m}^{\text{des}} \Gamma_{i,m} + \sum_{\alpha} \nu_{i\alpha} r_{\alpha,m}^{\text{surf}} - k_{i,m}^{\text{foul}} \Gamma_{i,m}. \quad (34)$$

Here $\Gamma_{i,m}$ is surface loading and θ_m is fractional site coverage. Surface poisoning or refractory organic buildup should be represented as loss of available mineral sites.

4.5 Coacervate droplets as the first evolvable compartments

A coacervate is not a protocell, but it is the first mobile compartment that can carry a partially heritable internal composition. Coacervates form when the pore/surface chemistry produces enough oppositely charged or mutually phase-separating polymers, peptides, hetero-oligomers, or metal-organic complexes.

In this model, a coacervate means a membrane-free, organic-rich liquid droplet inside water. It is a dense liquid phase, not a lipid vesicle. The reason it is important is that some molecules can become much more concentrated inside the droplet than in the surrounding water. This concentration can make otherwise slow reactions fast enough to matter.

Define a local coacervation score:

$$\Phi_{\text{coa}}(j) = \sum_{a \in A} \sum_{b \in B} \alpha_{ab} C_{a,j}^{\text{env}} C_{b,j}^{\text{env}} f_{\text{pH}}(\text{pH}_j) f_I(I_j). \quad (35)$$

The symbols in this score mean:

- $\Phi_{\text{coa}}(j)$ is the tendency of spatial cell j to form coacervate droplets.
- j labels one slice of the one-dimensional porous vent wall.
- A and B are two classes of molecules that can phase-separate together, for example oppositely charged peptides or organic polymers.
- $C_{a,j}^{\text{env}}$ is the concentration of molecule or pool a in the external water of cell j .
- $C_{b,j}^{\text{env}}$ is the concentration of molecule or pool b in the same external water.
- α_{ab} measures how strongly species class a and species class b promote droplet formation.
- $f_{\text{pH}}(\text{pH}_j)$ reduces or enhances coacervation depending on the local pH.
- $f_I(I_j)$ reduces or enhances coacervation depending on ionic strength. High salt can screen charges and destabilize some coacervates.

Coacervate birth is allowed when

$$\Phi_{\text{coa}}(j) > \Phi_{\text{coa}}^{\text{crit}}. \quad (36)$$

Here $\Phi_{\text{coa}}^{\text{crit}}$ is the threshold above which the local mixture is treated as able to phase-separate into droplets.

At birth, droplet d samples its internal inventory from the local environment with partition coefficients:

$$N_{i,d}(t_{\text{birth}}) \sim \text{Poisson} [K_{i,d}^{\text{coa}} C_{i,j}^{\text{env}} V_d N_A] . \quad (37)$$

The symbols in the birth equation mean:

- $N_{i,d}$ is the number of molecules of species i inside droplet d .
- t_{birth} is the time at which the droplet forms.
- $K_{i,d}^{\text{coa}}$ is the partition coefficient of species i into droplet d . If $K_{i,d}^{\text{coa}} > 1$, the molecule is enriched inside the droplet. If $K_{i,d}^{\text{coa}} < 1$, it is excluded.
- $C_{i,j}^{\text{env}}$ is the concentration of species i in the surrounding water of cell j .
- V_d is the droplet volume.
- N_A is Avogadro's number, converting concentration times volume into expected molecule number.
- $\text{Poisson}[\dots]$ means the actual molecule count is randomly sampled around the expected count.

The coacervate internal dynamics are

$$\frac{dC_{i,d}^{\text{coa}}}{dt} = \sum_{\alpha} \nu_{i\alpha} r_{\alpha,d}^{\text{coa}} + \frac{K_{i,d}^{\text{coa}} C_{i,j}^{\text{env}} - C_{i,d}^{\text{coa}}}{\tau_{i,d}^{\text{part}}} - k_{i,d}^{\text{loss}} C_{i,d}^{\text{coa}} . \quad (38)$$

The symbols in the coacervate dynamics equation mean:

- $C_{i,d}^{\text{coa}}$ is the concentration of species i inside coacervate droplet d .
- $\nu_{i\alpha}$ is the stoichiometric change of species i in reaction α : positive if the reaction produces i , negative if it consumes i .
- $r_{\alpha,d}^{\text{coa}}$ is the rate of reaction α inside droplet d .
- $K_{i,d}^{\text{coa}} C_{i,j}^{\text{env}}$ is the concentration that species i would approach inside the droplet if partitioning reached equilibrium.
- $\tau_{i,d}^{\text{part}}$ is the exchange or partitioning timescale. Large values mean slow exchange with the environment; small values mean fast equilibration.
- $k_{i,d}^{\text{loss}}$ is the loss or degradation rate of species i inside droplet d .

The partition coefficients K_i^{coa} and exchange times τ_i^{part} are evolvable or sampled traits in later simulations.

4.6 Coacervate to vesicle transition

The vesicle/protocell stage should be created by amphiphile accumulation, not assumed at initialization. Track amphiphile coverage on the coacervate interface:

$$\frac{dN_{L,d}}{dt} = k_{\text{ads}}A_dC_{L,j}^{\text{env}} - k_{\text{des}}N_{L,d} + R_{L,d}^{\text{internal}}. \quad (39)$$

The required number of amphiphiles for a membrane-like coating is

$$N_{L,d}^{\text{req}} \approx \frac{8\pi R_d^2}{a_L}, \quad (40)$$

where a_L is the molecular area per amphiphile. Define coverage

$$\theta_d = \frac{N_{L,d}}{N_{L,d}^{\text{req}}}. \quad (41)$$

When

$$\theta_d \geq \theta_{\text{ves}}, \quad (42)$$

the transition is

$$\text{coacervate droplet} \rightarrow \text{amphiphile-stabilized vesicle/protocell candidate}. \quad (43)$$

4.7 Lineage evolution should begin at the coacervate stage

For hydrothermal vents, selection should not start from fully formed protocells. The first selectable objects are coacervates or amphiphile-stabilized droplets that persist, concentrate feedstocks, and enhance productive chemistry. Their evolvable traits can be represented as

$$\Theta_d = \{K_i^{\text{coa}}, \tau_i^{\text{part}}, \chi_{\text{eff}}, P_i, \mathcal{K}_{d\alpha}^{\text{cat}}, \theta_{\text{div}}\}. \quad (44)$$

Here $\mathcal{K}_{d\alpha}^{\text{cat}}$ is the droplet's catalytic enhancement profile over the reference reaction network. Mutation changes these traits between parent and daughter droplets.

An operational fitness for early hydrothermal compartment evolution is

$$F_d = w_R R_{\text{productive}} + w_A A_{\text{autocatalytic}} + w_E C_{\text{energy}} + w_L C_{\text{amphiphile}} + w_S t_{\text{survival}} - w_D D_{\text{dilution}} - w_H H_{\text{hydrolysis}} - w_F F_{\text{fouling}} \quad (45)$$

The first computational target is not “life”. It is persistent, compartmentalized, autocatalytic chemistry that can transition from mineral pore/surface concentration into coacervate or vesicle-like objects.

4.8 Minimal code objects

The corresponding first implementation should expose these objects:

HydrothermalVent

```

spatial_cells: list[VentCell]
pore_compartments: list[PoreCompartment]
coacervates: list[CoacervateDroplet]
vesicles: list[VesicleCandidate]
```

VentCell

x, T, P, pH, ionic_strength, pe, flow_velocity
env_concentrations
mineral_surfaces

PoreCompartment

cell_id, radius, length, volume, surface_area
mineral_surface_area
concentrations
residence_time

CoacervateDroplet

cell_id, radius, volume
internal_counts
partition_coefficients
partition_times
catalytic_profile

VesicleCandidate

cell_id, radius, volume, area
internal_counts
amphiphile_count
permeability

The hydrothermal vent branch should therefore implement chemistry in this order:

1. parse the hydrothermal reference reaction network;
2. build environmental feedstock and mineral-surface fields;
3. run pore/surface chemistry;
4. allow coacervate birth from enriched polymer/charged-organic pools;
5. evolve coacervate internal chemistry and partitioning traits;
6. convert successful droplets into vesicle/protocell candidates.

5 Initial conditions for water worlds

The following numbers are not final observational constraints. They are scenario templates for simulation design. Europa and Ganymede ocean compositions are uncertain; published models commonly explore chloride-rich, sulfate-rich, carbonate-bearing, and water-rock alteration scenarios [7, 8, 9, 10]. The values below should therefore be treated as tunable hypotheses.

5.1 Scenario A: Europa-like oxidant-influenced chloride/sulfate ocean

Quantity	Symbol	Starting value / range
----------	--------	------------------------

Ocean temperature	T_{ocean}	260–290 K
Vent temperature	T_{vent}	320–500 K
Seafloor pressure	P_{sf}	50–150 MPa
pH initial guess	pH	5.5–9.5
Na	Na	0.05–0.70 mol/kgw
Cl	Cl	0.05–0.70 mol/kgw
Mg	Mg	10^{-3} –0.1 mol/kgw
Ca	Ca	10^{-4} –0.05 mol/kgw
Total carbonate	C(4)	10^{-4} – 10^{-1} mol/kgw
Sulfate	S(6)	10^{-5} – 10^{-1} mol/kgw
Sulfide	S(-2)	10^{-8} – 10^{-3} mol/kgw
Fe	Fe(2)	10^{-9} – 10^{-4} mol/kgw
Ni	Ni(2)	10^{-10} – 10^{-6} mol/kgw
Ammonia/ammonium	N(-3)	10^{-8} – 10^{-4} mol/kgw
Phosphate	P	10^{-9} – 10^{-5} mol/kgw
Silica	Si	10^{-5} – 10^{-2} mol/kgw
Hydrogen	H ₂ (aq)	10^{-8} – 10^{-2} mol/kgw
Methane	CH ₄ (aq)	10^{-9} – 10^{-3} mol/kgw

5.2 Scenario B: Ganymede-like more reduced ocean

Quantity	Symbol	Starting value / range
Ocean temperature	T_{ocean}	250–290 K
Vent temperature	T_{vent}	310–450 K
Seafloor pressure	P_{sf}	80–250 MPa
pH initial guess	pH	7.0–11.5
Na	Na	0.01–0.50 mol/kgw
Cl	Cl	0.01–0.50 mol/kgw
Mg	Mg	10^{-4} –0.05 mol/kgw
Ca	Ca	10^{-5} –0.02 mol/kgw
Total carbonate	C(4)	10^{-5} – 10^{-1} mol/kgw
Sulfate	S(6)	10^{-8} – 10^{-3} mol/kgw
Sulfide	S(-2)	10^{-7} – 10^{-2} mol/kgw
Fe	Fe(2)	10^{-9} – 10^{-4} mol/kgw
Ni	Ni(2)	10^{-10} – 10^{-6} mol/kgw
Ammonia/ammonium	N(-3)	10^{-8} – 10^{-4} mol/kgw
Phosphate	P	10^{-10} – 10^{-5} mol/kgw
Silica	Si	10^{-5} – 10^{-2} mol/kgw
Hydrogen	H ₂ (aq)	10^{-7} – 10^{-2} mol/kgw
Methane	CH ₄ (aq)	10^{-8} – 10^{-3} mol/kgw

5.3 Scenario C: generic Enceladus-like alkaline hydrothermal ocean

This case is useful as a positive-control scenario for alkaline hydrothermal chemistry.

Quantity	Symbol	Starting value / range
----------	--------	------------------------

Ocean temperature	T_{ocean}	270–300 K
Vent temperature	T_{vent}	330–500 K
Seafloor pressure	P_{sf}	5–50 MPa
pH initial guess	pH	8.5–12.0
Na	Na	0.05–0.80 mol/kgw
Cl	Cl	0.05–0.80 mol/kgw
Carbonate	C(4)	10^{-4} – 10^{-1} mol/kgw
Sulfide	S(-2)	10^{-7} – 10^{-2} mol/kgw
Hydrogen	H2(aq)	10^{-6} – 10^{-1} mol/kgw
Methane	CH4(aq)	10^{-8} – 10^{-2} mol/kgw
Silica	Si	10^{-5} – 10^{-2} mol/kgw

6 PHREEQC input templates

6.1 Ocean boundary template

```

TITLE Europa-like ocean boundary
SOLUTION 1 Ocean
  temp      {T_ocean_C}
  pressure  {P_ocean_atm}
  units     mol/kgw
  pH        {pH_guess} charge
  pe        {pe_guess}
  Na        {Na}
  Cl        {Cl}
  Mg        {Mg}
  Ca        {Ca}
  C(4)      {C_total}
  S(6)      {Sulfate}
  S(-2)     {Sulfide}
  Fe(2)     {Fe2}
  Ni        {Ni2}
  N(-3)     {NH3_total}
  P         {P_total}
  Si        {Si_total}
  H(0)      {H2_total}
END

```

6.2 Vent boundary template

```

TITLE Reduced hydrothermal vent boundary
SOLUTION 2 Vent
  temp      {T_vent_C}
  pressure  {P_vent_atm}
  units     mol/kgw
  pH        {pH_vent_guess} charge
  pe        {pe_vent_guess}
  Na        {Na_vent}

```

```

Cl      {Cl_vent}
Mg      {Mg_vent}
Ca      {Ca_vent}
C(4)    {C_vent}
S(-2)   {Sulfide_vent}
Fe(2)   {Fe2_vent}
Ni      {Ni2_vent}
N(-3)   {NH3_vent}
P       {P_vent}
Si      {Si_vent}
H(0)    {H2_vent}
EQUILIBRIUM_PHASES
Mackinawite  0.0  0.0
Pyrite       0.0  0.0
Magnetite    0.0  0.0
Brucite      0.0  0.0
Calcite      0.0  0.0
END

```

7 PHREEQC outputs required by the kinetic network

At each grid cell, extract:

Output	Symbol	Use in kinetic network
pH	pH	Acid/base control, vesicle stability
pe/Eh	pe or E_h	Redox availability
Ionic strength	I	Activity corrections, vesicle stress
CO ₂ (aq)	a_{CO_2}	Carbon fixation
HCO ₃ ⁻	$a_{\text{HCO}_3^-}$	Carbonate buffer, carbon source
CO ₃ ²⁻	$a_{\text{CO}_3^{2-}}$	Carbonate precipitation
H ₂ (aq)	a_{H_2}	Electron donor
CO(aq)	a_{CO}	Carbonyl chemistry
H ₂ S, HS ⁻	a_{sulfide}	FeS formation, thioester chemistry
Fe ²⁺ , Ni ²⁺	$a_{\text{Fe}}, a_{\text{Ni}}$	Catalytic mineral availability
NH ₃ , NH ₄ ⁺	a_{N}	Amino acid precursors
Phosphate species	a_{Pi}	Acetyl phosphate, activation chemistry
Mg ²⁺ , Ca ²⁺	$a_{\text{Mg}}, a_{\text{Ca}}$	Fatty acid precipitation/stability
Mineral SI	SI_m	Surface availability and precipitation sinks

8 Reaction network

The following reaction network is deliberately written as an implementable starting model. Some reactions are elementary-looking; others are lumped effective reactions representing families of pathways. Each reaction should carry a confidence flag and a literature source in the code.

8.1 Module A: aqueous acid–base and geochemical equilibria

These reactions should mostly be handled by PHREEQC, not by the custom kinetic layer.

ID	Reaction	Role
G1	$\text{CO}_2(\text{aq}) + \text{H}_2\text{O} \rightleftharpoons \text{HCO}_3^- + \text{H}^+$	carbonate speciation
G2	$\text{HCO}_3^- \rightleftharpoons \text{CO}_3^{2-} + \text{H}^+$	carbonate speciation
G3	$\text{H}_2\text{S} \rightleftharpoons \text{HS}^- + \text{H}^+$	sulfide speciation
G4	$\text{HS}^- \rightleftharpoons \text{S}^{2-} + \text{H}^+$	sulfide speciation
G5	$\text{NH}_4^+ \rightleftharpoons \text{NH}_3 + \text{H}^+$	nitrogen speciation
G6	$\text{H}_3\text{PO}_4 \rightleftharpoons \text{H}_2\text{PO}_4^- + \text{H}^+$	phosphate speciation
G7	$\text{H}_2\text{PO}_4^- \rightleftharpoons \text{HPO}_4^{2-} + \text{H}^+$	phosphate speciation
G8	$\text{HPO}_4^{2-} \rightleftharpoons \text{PO}_4^{3-} + \text{H}^+$	phosphate speciation
G9	$\text{Fe}^{2+} + \text{HS}^- \rightarrow \text{FeS}(\text{s}) + \text{H}^+$	FeS precipitation
G10	$\text{Ni}^{2+} + \text{HS}^- \rightarrow \text{NiS}(\text{s}) + \text{H}^+$	NiS precipitation
G11	$\text{Ca}^{2+} + \text{CO}_3^{2-} \rightarrow \text{CaCO}_3(\text{s})$	carbonate sink
G12	$\text{Mg}^{2+} + 2\text{OH}^- \rightarrow \text{Mg}(\text{OH})_2(\text{s})$	brucite sink
G13	$3\text{Fe}^{2+} + 4\text{H}_2\text{O} \rightarrow \text{Fe}_3\text{O}_4(\text{s}) + 8\text{H}^+ + 2\text{e}^-$	magnetite/redox
G14	$\text{FeS}(\text{s}) + \text{S}^0 \rightarrow \text{FeS}_2(\text{s})$	pyrite formation
G15	$\text{SiO}_2(\text{aq}) \rightarrow \text{SiO}_2(\text{s})$	silica precipitation

8.2 Module B: C1 carbon fixation and redox entry

ID	Reaction	Catalyst / control
R1	$\text{CO}_2 + \text{H}_2 \rightarrow \text{HCOO}^- + \text{H}^+$	FeS/NiS/magnetite
R2	$\text{CO}_2 + \text{H}_2 \rightarrow \text{CO} + \text{H}_2\text{O}$	Fe/Ni minerals
R3	$\text{CO} + \text{H}_2\text{O} \rightarrow \text{CO}_2 + \text{H}_2$	water–gas shift
R4	$\text{CO}_2 + 4\text{H}_2 \rightarrow \text{CH}_4 + 2\text{H}_2\text{O}$	methanation sink
R5	$\text{HCOO}^- + \text{H}^+ \rightleftharpoons \text{HCOOH}$	PHREEQC/pH
R6	$\text{CO}_2 + \text{HS}^- + \text{H}^+ \rightarrow \text{COS} + \text{H}_2\text{O}$	sulfide activation
R7	$\text{COS} + \text{H}_2\text{O} \rightarrow \text{CO}_2 + \text{H}_2\text{S}$	hydrolysis sink
R8	$\text{COS} + \text{NH}_3 \rightarrow \text{NH}_2\text{CO}^- \text{SH}$	carbamoyl/thiocarbamate pool
R9	$\text{CO} + \text{H}_2\text{S} \rightarrow \text{COS} + \text{H}_2$	sulfur-carbonyl pool
R10	$\text{HCOO}^- + \text{H}_2 \rightarrow \text{HCHO} + \text{OH}^-$	reduced C1 pool

8.3 Module C: thioester and acetyl-phosphate activation

ID	Reaction	Catalyst / control
R11	$\text{CO} + \text{CH}_3\text{SH}^- \rightarrow \text{CH}_3\text{COSR}$	FeS/NiS surface
R12	$\text{CO}_2 + \text{CH}_3\text{SH} + 2\text{H}_2^- \rightarrow \text{CH}_3\text{COSR} + 2\text{H}_2\text{O}$	Fe/Ni sulfides
R13	$\text{CH}_3\text{COSR} + \text{H}_2\text{O}^- \rightarrow \text{CH}_3\text{COO}^- + \text{RSH} + \text{H}^+$	hydrolysis sink
R14	$\text{CH}_3\text{COSR} + \text{Pi}^- \rightarrow \text{CH}_3\text{COOPO}_3^{2-} + \text{RSH}$	acetyl phosphate
R15	$\text{CH}_3\text{COOPO}_3^{2-} + \text{H}_2\text{O}^- \rightarrow \text{CH}_3\text{COO}^- + \text{Pi} + \text{H}^+$	hydrolysis sink
R16	$\text{CH}_3\text{COOPO}_3^{2-} + \text{NH}_3^- \rightarrow \text{acetamide} + \text{Pi}$	amide side branch
R17	$\text{CH}_3\text{COSR} + \text{NH}_3^- \rightarrow \text{thioacetamide} + \text{RSH}$	nitrogen side branch
R18	$\text{CH}_3\text{COSR} + \text{HCOO}^- \rightarrow \text{acetoacetyl_SR}$	C–C extension
R19	$\text{CH}_3\text{COSR} + \text{CO}_2 + \text{H}_2^- \rightarrow \text{pyruvate} + \text{RSH}$	C3 precursor
R20	$\text{CH}_3\text{COO}^- + \text{H}^+ \rightleftharpoons \text{CH}_3\text{COOH}$	PHREEQC/pH

8.4 Module D: C2–C5 central organic precursor network

ID	Reaction	Catalyst / control
R21	$\text{HCHO} + \text{HCHO}^- \rightarrow \text{glycolaldehyde}$	mineral/base catalysis
R22	$\text{glycolaldehyde} + \text{HCHO}^- \rightarrow \text{glyceraldehyde}$	sugar precursor
R23	$\text{glycolaldehyde} + \text{HCN}^- \rightarrow \text{glycolonitrile}$	nitrile branch
R24	$\text{HCHO} + \text{HCN}^- \rightarrow \text{glycolonitrile}$	nitrile branch
R25	$\text{glycolonitrile} + 2\text{H}_2\text{O}^- \rightarrow \text{glycolate} + \text{NH}_3$	hydrolysis
R26	$\text{acetate} + \text{CO}_2 + \text{H}_2^- \rightarrow \text{pyruvate} + \text{H}_2\text{O}$	Fe/Ni minerals
R27	$\text{pyruvate} + \text{CO}_2 + \text{H}_2^- \rightarrow \text{oxaloacetate}$	C4 precursor
R28	$\text{pyruvate} + \text{H}_2^- \rightarrow \text{lactate}$	reduction sink
R29	$\text{pyruvate}^- \rightarrow \text{acetate} + \text{CO}_2$	decarboxylation sink
R30	$\text{glyoxylate} + \text{acetate}^- \rightarrow \text{malate}$	C4 pool abstraction
R31	$\text{malate}^- \rightarrow \text{fumarate} + \text{H}_2\text{O}$	dehydration
R32	$\text{fumarate} + \text{H}_2^- \rightarrow \text{succinate}$	reduction
R33	$\text{succinate} + \text{CO}_2 + \text{H}_2^- \rightarrow \text{alpha-ketoglutarate}$	C5 precursor
R34	$\text{alpha-ketoglutarate}^- \rightarrow \text{succinate} + \text{CO}_2$	degradation sink
R35	$\text{oxaloacetate}^- \rightarrow \text{pyruvate} + \text{CO}_2$	degradation sink

8.5 Module E: amino-acid precursor network

ID	Reaction	Catalyst / control
R36	$\text{glyoxylate} + \text{NH}_3 + \text{H}_2^- \rightarrow \text{glycine} + \text{H}_2\text{O}$	reductive amination
R37	$\text{pyruvate} + \text{NH}_3 + \text{H}_2^- \rightarrow \text{alanine} + \text{H}_2\text{O}$	reductive amination

R38	oxaloacetate + NH3 + H2- > aspartate + H2O	reductive amination
R39	alpha-ketoglutarate + NH3 + H2- > glutamate + H2O	reductive amination
R40	HCHO + NH3 + HCN- > aminoacetonitrile + H2O	Strecker-like
R41	aminoacetonitrile + 2H2O- > glycine + NH3	hydrolysis
R42	acetaldehyde + NH3 + HCN- > alanine_nitrile + H2O	Strecker-like
R43	alanine_nitrile + 2H2O- > alanine + NH3	hydrolysis
R44	amino_acid + H2O- > degradation_products	hydrolysis/degradation
R45	amino_acid + mineral- > amino_acid_surface	retention

8.6 Module F: peptide and activated condensation network

ID	Reaction	Catalyst / control
R46	AA + acetylP- > AA* + acetate	activation
R47	AA* + AA- > peptide2 + Pi	condensation
R48	peptide _n + AA*- > peptide _{n+1} + Pi	chain growth
R49	peptide _n + H2O- > peptide _{n-1} + AA	hydrolysis
R50	peptide _n + mineral- > peptide _n _surface	retention/catalysis
R51	peptide_surface + FA- > peptide_FA_complex	membrane stabilization
R52	AA* + H2O- > AA + Pi	activation loss
R53	peptide _n - > degraded_organic_pool	thermal degradation

8.7 Module G: fatty-acid, amphiphile, and vesicle network

ID	Reaction	Catalyst / control
R54	CO + H2- > CH2* + H2O	Fischer-Tropsch-like
R55	CH2* + acyl_C _n - > acyl_C _{n+1}	chain growth
R56	acetyl-SR + CH2*- > acyl_C3-SR	chain initiation
R57	acyl_C _n -SR + H2O- > FA_C _n + RSH	fatty acid release
R58	FA_C _n + Mg ²⁺ - > Mg(FA) ₂ (s)	precipitation sink
R59	FA_C _n + Ca ²⁺ - > Ca(FA) ₂ (s)	precipitation sink
R60	FA_C _n + H+ <=> HFA_C _n	pH-dependent
R61	FA_C _n - > micelle	protonation
R62	micelle + FA_C _n - > vesicle	self-assembly
R63	vesicle + FA_C _n - > larger-vesicle	growth
R64	vesicle + peptide- > stabilized-vesicle	membrane growth
R65	vesicle- > micelle + FA_loss	stabilization disruption

9 Rate-law templates

For each reaction α , store:

```
{
  "id": "R1",
  "reactants": {"CO2": 1, "H2": 1},
  "products": {"formate": 1, "H+": 1},
  "phase": "surface_aqueous",
  "catalysts": ["FeS", "NiS", "magnetite"],
  "k0": null,
  "Ea_kJ_mol": null,
  "pH_window": [8.0, 11.5],
  "T_window_K": [273, 500],
  "depends_on_phreeqc": ["CO2", "H2", "pH", "pe", "FeS_SI", "NiS_SI"],
  "confidence": "medium",
  "reference": ["MartinRussell2006", "Sojo2016"]
}
```

Use activities where possible:

$$r_{\text{CO2} \rightarrow \text{formate}} = k_1(T) a_{\text{CO2}} a_{\text{H2}} h_{\text{FeS/NiS}} f_{\text{pH}} f_{\text{redox}}. \quad (46)$$

For fatty-acid vesicle formation:

$$r_{\text{ves}} = k_{\text{ves}} C_{\text{FA}} f_{\text{pH}}(\text{pH}) f_I(I) f_{\text{div}}(a_{\text{Mg}^{2+}}, a_{\text{Ca}^{2+}}), \quad (47)$$

with f_{div} decreasing when divalent cations precipitate fatty acids.

10 Numerical algorithm

1. Build planet object and compute $T(x)$, $P(x)$, flow velocity $u(x)$, and energy/redox fluxes.
2. Initialize bulk element totals for ocean and vent endmembers.
3. For each grid cell, mix ocean and vent endmembers:

$$b_i(x) = \lambda(x) b_i^{\text{vent}} + [1 - \lambda(x)] b_i^{\text{ocean}}. \quad (48)$$

4. Run PHREEQC at each grid cell using $T(x)$, $P(x)$, $b_i(x)$, gas constraints, and mineral phases.
5. Extract pH, pe, ionic strength, activities, and mineral saturation indices.
6. Evaluate kinetic rates r_α for the organic network using PHREEQC output.
7. Update organic concentrations by reaction-transport.

8. Feed selected organic totals back into PHREEQC only if they significantly affect charge balance, complexation, or precipitation.
9. Compute local protocell nucleation probability.
10. Create, evolve, divide, or destroy protocell agents.

11 Python object interfaces

11.1 Planet state

```
@dataclass
class PlanetState:
    name: str
    radius_m: float
    mass_kg: float
    gravity_m_s2: float
    ocean_depth_m: float
    ice_shell_thickness_m: float
    tidal_heat_flux_W_m2: float
    radiogenic_heat_flux_W_m2: float
    oxidant_flux_mol_m2_s: float
    rock_model: str
```

11.2 Geochemical cell input

```
@dataclass
class GeoCellInput:
    T_K: float
    P_Pa: float
    totals_mol_kgw: dict
    pH_guess: float
    pe_guess: float
    minerals: list
    gases: dict
```

11.3 PHREEQC output state

```
@dataclass
class GeoState:
    pH: float
    pe: float
    ionic_strength: float
    activities: dict
    concentrations: dict
    saturation_indices: dict
    mineral_amounts: dict
```

11.4 Organic network state

```
@dataclass
class OrganicState:
    formate: float
    formaldehyde: float
    CO: float
    acetate: float
    pyruvate: float
    oxaloacetate: float
    acetyl_thioester: float
    acetyl_phosphate: float
    amino_acid_pool: float
    activated_amino_acid_pool: float
    peptide_pool: float
    fatty_acid_pool: float
    micelle_pool: float
    vesicle_pool: float
```

11.5 Protocell agent

```
@dataclass
class Protocell:
    position: float
    lipid_inventory: float
    peptide_inventory: float
    energy_inventory: float
    permeability: float
    stability: float
    catalyst_genotype: dict
    size: float
    age: float
```

12 References

References

- [1] D. L. Parkhurst, *User's guide to PHREEQC: A computer program for speciation, reaction-path, advective-transport, and inverse geochemical calculations*, U.S. Geological Survey Water-Resources Investigations Report 95-4227, 1995. <https://pubs.usgs.gov/publication/wri954227>
- [2] D. L. Parkhurst and C. A. J. Appelo, *User's guide to PHREEQC (Version 2): A computer program for speciation, batch-reaction, one-dimensional transport, and inverse geochemical calculations*, U.S. Geological Survey Water-Resources Investigations Report 99-4259, 1999. <https://pubs.usgs.gov/publication/wri994259>

- [3] D. L. Parkhurst and C. A. J. Appelo, *Description of input and examples for PHREEQC version 3*, U.S. Geological Survey Techniques and Methods, 2013. <https://water.usgs.gov/water-resources/software/PHREEQC/>
- [4] W. Martin and M. J. Russell, On the origin of biochemistry at an alkaline hydrothermal vent, *Philosophical Transactions of the Royal Society B*, 2007. <https://pmc.ncbi.nlm.nih.gov/articles/PMC2442388/>
- [5] V. Sojo, B. Herschy, A. Whicher, E. Camprubi, and N. Lane, The origin of life in alkaline hydrothermal vents, *Astrobiology*, 2016. <https://pubmed.ncbi.nlm.nih.gov/26841066/>
- [6] A. Whicher et al., Acetyl phosphate as a primordial energy currency at the origin of life, *Origins of Life and Evolution of Biospheres*, 2018. <https://pmc.ncbi.nlm.nih.gov/articles/PMC6061221/>
- [7] S. Tan et al., Laboratory experiments on sulfate reduction at 100 MPa: Implications for Europa’s ocean, *Icarus*, 2021. <https://www.sciencedirect.com/science/article/abs/pii/S0019103520305522>
- [8] S. P. Thompson et al., Laboratory exploration of mineral precipitates from Europa’s ocean composition analogs, 2021. <https://pmc.ncbi.nlm.nih.gov/articles/PMC8493616/>
- [9] NASA, Europa’s ocean may have an Earthlike chemical balance, 2016. <https://science.nasa.gov/missions/europa-clipper/europas-ocean-may-have-an-earthlike-chemical-balance/>
- [10] P. Gavin and S. Vance, Modeling hydrothermal vents on Europa, Lunar and Planetary Science Conference, 2012. <https://www.lpi.usra.edu/meetings/lpsc2012/pdf/1683.pdf>